MILITARY SPECIFICATION

MOTOR, DIRECT CURRENT FOR CONTROL-MONITORS DCU-116/A, DCU-117/A, AND DCU-121/A

1. SCOPE. This specification covers the requirements for design, manufacture and testing for one type of miniature direct current motor for the Control-Monitors, DCU-116/A, DCU-117/A and DCU-121/A.

2. APPLICABLE DOCUMENTS. The following documents, of the issue in effect on the date of invitation for bids, form a part of this specification to the extent specified herein:

SPECIFICATIONS

Military

MIL-D-1000	Drawings, Engineering and Associated Lists
MIL-I-8500	Interchangeability and Replaceability, Physical, of Component Parts for Aircraft
MIL-E-17555	Electronic and Electrical Equipment and Associated Repair Parts, Preparation for Delivery Of

STANDARDS

Military

MIL-STD-16	Electric and Electronic Reference Designations
MIL-STD-129	Marking for Shipment and Storage
MIL-STD-130	Identification Marking of U.S. Military Property
MIL-STD-810	Environmental Test Methods for Aerospace and Ground Equipment
MIL-STD-831	Test Reports, Preparation Of
MIL-STD-785	Requirements for Reliability Program for Systems and Equipment

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DRAWINGS

Military

66D13206	Motor, Direct Current for Control-Monitors

(Copies of the above documents which are required by contractors in connection with specific procurement functions should be obtained from the Administrative Contracting Officer or as directed by the procuring activity.)

3. REQUIREMENTS

3.1 Preproduction. This specification makes provisions for preproduction testing.

3.2 Design and Construction. The motor shall be designed and constructed in conformance to Drawing No. 66D13206.

3.2.1 Interchangeability. Parts functionally and dimensionally interchangeable shall be identified by the same part number. When parts are not interchangeable, the criteria of MIL-D-1000 shall apply and new drawings shall be prepared as required. Physical interchangeability shall be in accordance with MIL-I-8500.

3.3 Performance. Satisfactory performance of the motor shall consist of demonstration that the motor meets the tests specified herein.

3.3.1 Reliability. The contractor shall initiate and maintain a reliability program in accordance with MIL-STD-785. This program shall be subject to review and modification by the procuring activity.

3.3.2 Low Voltage. The motor shall be capable of being subjected to an input voltage of 23.75 volts DC and an input current not to exceed 300 milliamperes.

3.3.3 High Voltage. The motor shall be capable of being subjected to an input voltage of 32.25 volts DC and an input current not to exceed 300 milliamperes.

3.3.4 Coast Angle. The motor shall have the capability of the output shaft stopping in less than 30 degrees after the removal of voltage.

3.3.5 Rotation. The motor shall have the capability of rotating counterclockwise.

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3.3.6 Leakage Current. The leakage current between the motor leads and the motor case shall be 5 milliamperes with a potential of 500 ± 25 volts DC.

3.3.7 Surge Current. The motor shall be capable of being subjected to the surge current test as specified in the specification.

3.4 Environmental Requirements. The motor shall be capable of satisfactory performance during or after, as applicable, subjection to the following environmental conditions:

a. High Temperature. The motor shall be subjected to a temperature of +160°F.

b. Low Temperature. The motor shall be subjected to a temperature of -65°F.

c. Temperature Shock incident to service use.

d. Low Pressure as encountered in flight.

e. Vibration incident to transportation and handling.

f. Shock. The motor shall be capable of being subjected to a total of 18 impact shocks of 100g.

g. Acceleration. The motor will be capable of being subjected to a constant acceleration of 100g for at least 60 seconds applied in each direction along each of three mutually perpendicular axes.

h. Fungus such as encountered in storage or use in tropical climates.

3.4.1 Vibration Cycling at +160°F. The motor shall be capable of being vibrated while at +160°F.

3.4.2 Vibration Cycling at -65°F. The motor shall be capable of being vibrated while at -65°F.

3.4.3 Shock at -65°F. The motor shall be capable of being shocked while at -65°F.

3.4.4 Shock at +160°F. The motor shall be capable of being shocked while at +160°F.

3.4.5 Acceleration at +160°F. The motor shall be capable of being accelerated while at +160°F.

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3.4.6 Acceleration at -65°F. The motor shall be capable of being accelerated while at -65°F.

3.4.7 Life Test. The motor shall be capable of being subjected to the life test as specified in the specification.

3.4.8 Temperature Storage. The motor shall be capable of being subjected to a temperature of +160°F for a period of 100 hours.

3.4.9 Identification of Product. The motor and all parts shall be marked in accordance with MIL-STD-130.

4. QUALITY ASSURANCE

4.1 Classification of Tests. Inspection and testing of the motor to this specification shall consist of:

a. Preproduction Tests

b. Acceptance Tests

c. Functional Tests

4.2 Preproduction Testing

4.2.1 Preproduction Test Samples. The preproduction test samples shall be representative of the production motor. They shall be tested at a laboratory designated by the procuring activity, or, when so stated in the contract, at the contractor's plant, under the supervision of the procuring activity.

4.2.1.1 Preproduction Test Reports. In the event that preproduction testing is performed at the contractor's plant, the contractor shall, after the completion of the preproduction tests, prepare a preproduction test report in accordance with MIL-STD-831 and furnish three complete copies of the report to the procuring activity.

4.2.2 Preproduction Tests. Preproduction tests shall consist of all tests required under "4.5 Test Methods" as listed in Table 1 and in the order specified.

4.3 Acceptance Tests. Acceptance tests shall consist of the following tests:

a. Individual Tests

b. Sampling Tests

4.3.1 Individual Tests. Each motor shall be subjected to the individual tests listed in Table 1 in the order specified.

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TABLE I

TEST	REQUIREMENT PARAGRAPH	TEST PARAGRAPH	PREPRO DUCTION TEST	SAMPLING TEST	INDIVIDUAL TEST
Examination of Product	3.2	4.5.1	X		
Leakage Current	3.3.6	4.5.4	X		X
Part I	3.3.6	4.5.4.1	X		X
Part II	3.3.6	4.5.4.2	X		
* Rotation	3.3.5	4.5.3.4	X		X
Low Voltage	3.3.2	4.5.3.1	X		X
High Voltage	3.3.3	4.5.3.2	X		X
Coast Angle	3.3.4	4.5.3.3	X		X
*Temperature Shock	3.4.(c)	4.5.5.3	X		
*Low Pressure and Temperature	3.4.(d)	4.5.5.4	X		
Acceleration	3.4.(g)	4.5.5.7	X		
*Shock	3.4.(f)	4.5.5.6	X		
*Fungus	3.4.(h)	4.5.5.8	X		
Surge Current	3.3.7	4.5.4.3	X		
*Vibration Cycling at +160°F	3.4.1	4.5.7	X		
*Vibration Cycling at -65°F	3.4.2	4.5.8	X		
*Shock at +160°F	3.4.4	4.5.10	X		
*Shock at -65°F	3.4.3	4.5.9	X		
Acceleration at +160°F	3.4.5	4.5.11	X		
Acceleration at -65°F	3.4.6	4.5.12	X		
*Vibration	3.4.(e)	4.5.5.5	X	X	
Low Temperature	3.4.(b)	4.5.5.1	X	X	
High Temperature	3.4.(a)	4.5.5.2	X	X	
High Temperature Storage	3.4.8	4.5.13	X	X	
*Life Test	3.4.7	4.5.6	X	X	

*This test is destructive or degrading.

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4.3.2 Sampling Plan and Tests

4.3.2.1 Lot Size and Number of Samples. The quantity of samples shall be selected according to lot size.

Lot Size	Number of Samples
50-100	6
101-200	10
201-300	16
301-400	20
401-500	24
Over 500	28

4.3.2.2 Sampling Plan. The number of samples selected at random from each lot shall be categorized into three groups, Group I, Group II and Group III. Group I testing shall consist of the low temperature and high temperature tests and shall be performed on all samples. Group II testing shall consist of the high temperature storage and vibration sinusoidal cycling tests which will be performed on half of the sample units randomly selected from those units used for Group I testing. Group III testing shall consist of the life test which will be performed on the sample units remaining from Group I after selection of Group II. Prior to subjection to the sampling tests, the sample motors shall have passed the individual tests.

4.3.2.3 Sampling Tests. Sampling tests shall consist of all the tests shown in Table 1 in the order specified.

4.3.3 Rejection and Retest. When the sample motor fails to meet the specified test, no motor on hand or later produced shall be accepted until the extent and cause of failure are determined. The failure shall be corrected and the unit shall demonstrate compliance by satisfactorily passing the failed test.

4.3.4 Defects in Items Already Accepted. The investigation of a test failure could indicate that defects may exist in items already accepted. If so, the contractor shall fully advise the procuring activity of all defects likely to be found and methods of correcting them.

4.4 Test Conditions. In addition to those conditions listed below, test conditions shall be specified in each specific test under "4.5 Test Methods."

Tolerances. The limiting values specified herein are absolute. Unless otherwise specified, tolerances shall be:

DC Voltages	± 25 v
Torque	± 10 percent

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Barometric Pressure	± 0.2 inch of Hg
Temperature	± 10°F
Time	± 5 percent
Acceleration	± 5 percent
Vibration	
Acceleration or Amplitude	± 15 percent
Frequency	± 2 percent or 3 cps whichever is greater
Shock	
Amplitude	± 15 percent
Duration	± 15 percent

4.4.1 Test Equipment. The equipment for measuring motor shaft speed shall be accurate to within ± 0.3 percent.

4.4.2 Power Source. The power source (including power supply and external circuitry) shall have the following characteristics:

a. Regulation. Under steady-state conditions, the source shall provide a regulated voltage at the motor terminals of ± 200 millivolts from no-load to 10 amperes.

b. Transient Response. The source shall recover to within 1 percent of the regulated voltage within 100 microseconds following a load step increase from 0 to 10 amperes.

4.5 Test Methods

4.5.1 Examination of Product. Each motor shall be examined visually to determine compliance with all requirements to this specification for which tests are not specified. Failure to comply with all requirements shall be cause for rejection.

4.5.2 Conformance to Drawings. Each motor shall be inspected to determine conformance to the engineering drawings, subsidiary drawings and specifications and standards referenced therein. Failure to conform to these criteria shall be cause for rejection.

4.5.3 Function Tests. All measurement of motor performance shall be taken within 20 seconds after power is applied. The 360 ohm resistor shown on drawing 66D13206 shall be used for all testing of the motor.

4.5.3.1 Low Voltage. Apply an input voltage of 23.75 volts DC to the motor with a torque load of 20 ounce-inches. The steady-state output shaft speed shall be 15 rpm ± 1.5 rpm. The steady-state, input current shall not exceed 300 milliamperes.

4.5.3.2 High Voltage. Apply an input voltage of 32.25 volts DC to the motor with a load of less than 5 oz.-in. on the output shaft. The steady-state output shaft speed shall be 15 rpm ± 1.5 rpm. The input current shall not exceed 300 milliamperes (ma).

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4.5.3.3 Coast Angle. With an input voltage of 32.25 volts DC applied to the motor and with no load on the motor shaft, the output shaft shall stop in less than 30 degrees after removal of voltage.

4.5.3.4 Rotation. With the red lead of the motor connected to the positive terminal of the power supply and the black lead connected to the negative terminal, the direction of rotation of the output shall be counterclockwise when viewed from the outer end of the shaft.

4.5.4 Leakage Current

4.5.4.1 Part I. Initial Testing. The motor will be tested prior to any test designated as destructive or degrading, and prior to acceptance of the motor by the procuring activity, the leakage current measured shall not exceed 5 milliamperes when a potential of 500 ± 25 volts DC is applied between the motor leads (all three leads shorted together) and the motor case for a period of 30 seconds minimum.

4.5.4.2 Part II. Subsequent Testing. Upon testing designated as destructive or degrading, and following acceptance of the motor by the procuring activity the leakage current measured shall not exceed 10 milliamperes when a potential of 100 ± 10 volts DC is applied between the motor leads (all three leads shorted together) and the motor case for a period of 30 seconds minimum.

4.5.4.3 Surge Current. Apply 32.25 volts DC to the motor and determine the current surge at no load. The peak current shall be less than 3.5 amperes and shall decay to 500 milliamperes or less within 350 milliseconds after power is applied.

4.5.5 Environmental Tests. During or after subjection to the environmental tests, as specified in the test, the motor shall be subjected to the functional tests. Failure to satisfy the requirements of the functional tests shall be considered as failure to pass the environmental test and cause for rejection.

4.5.5.1 Low Temperature. The low temperature test shall be performed in accordance with Procedure 1, Method 502.1, MIL-STD-810. The motor shall be subjected to a temperature of -65°F for a period of 3 hours. With the motor stabilized at -65°F, the motor shall be tested to and meet the requirements of this specification, except that the current limit shall be 350 milliamperes. The motor shall then be allowed to stabilize at room temperature and shall then meet the requirements of this specification.

4.5.5.2 High Temperature. The high temperature test shall be performed in accordance with Procedure 1, Method 501.1, MIL-STD-810. The motor shall be subjected to a temperature of + 160°F for a period

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of 2 hours. With the motor stabilized at +160°F, the motor shall be tested to and meet the requirements of this specification. The motor shall then be allowed to stabilize at room temperature and shall then meet the requirements of the low voltage, high voltage, and coast angle tests of this specification.

4.5.5.3 Temperature Shock. The temperature shock test shall be performed in accordance with Procedure 1, Method 503.1, MIL-STD-810. The motor shall be tested without thermal jackets. The motor shall be subjected to a temperature of 160°F for a minimum of 2 hours. The motor shall be removed from the +160°F chamber and within 5 minutes shall be placed in another chamber where temperature is maintained at -65°F. After being subjected to -65°F for a minimum of 3 hours, the motor shall then be removed from the -65°F chamber and within 5 minutes placed in the chamber whose temperature is maintained at +160°F. After remaining in this chamber for a minimum of five hours, the motor shall be removed, allowed to stabilize at room temperature, and shall meet the requirements of this specification.

4.5.5.4 Low Pressure. The pressure test shall be performed in accordance with Procedure II, Method 500.1, MIL-STD-810, except that after two hours the internal pressure of the chamber shall be reduced to 2.0 ± 0.1 inches of mercury, absolute. The motor shall be exposed to these conditions for one hour. At the conclusion of this period and while still at the specified pressure and temperature conditions, the motor shall meet the high voltage requirements of this specification, except that the steady-state current shall not exceed 350 milliamperes. The motor shall be allowed to stabilize at room pressure and temperature and shall then meet the requirements of the low voltage, high voltage, and coast angle tests of this specification.

4.5.5.5 Vibration. Vibration tests of the motor shall be performed in accordance with MIL-STD-810, Equipment Class 1, Mounting A, Figure 514.1, Test Curve C, except that in the vibration sinusoidal cycling the motor shall be vibrated successively along each of the three perpendicular axes. The frequency of vibration shall be varied from 10-2000-10 cps using a sweep of 30 minutes duration. Three sweeps shall be accomplished along each axis with input acceleration as specified in MIL-STD-810, Method 514.1, Curve C. The motor shall satisfy the requirements of the low voltage, high voltage, coast angle tests and leakage current, Part II tests at the conclusion of sinusoidal cycling and at the conclusion of resonance dwell.

4.5.5.6 Shock. The shock test of the motor shall be performed in accordance with MIL-STD-810, Method 516.1, Procedure V. After completion of the shock testing, the motor shall meet the requirements of the low voltage, high voltage, coast angle, and leakage current, Part II tests of this specification.

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4.5.5.7 Acceleration. The acceleration test shall be performed in accordance with MIL-STD-810, Method 513.1, Procedure II. The motor shall be subjected to a constant acceleration of 100g for at least 60 seconds applied in each direction along each of three mutually perpendicular axes. The motor shall meet the requirements of test during each acceleration period, except that the current value may be up to 3000 milliamperes and speed may be 15.0 + 1.5, -2.0 rpm. The motor shall meet the requirements of the low voltage, high voltage, coast angle, and leakage current, Part II tests of this specification after each acceleration period.

4.5.5.8 Fungus. The fungus test shall be performed in accordance with MIL-STD-810, Method 507.1, Procedure 1. The motor shall be disassembled, finger prints removed, and the lubricants shall not be removed from the bearings and gears prior to the test. At the end of the test cycle, if more than 4 percent of the surface of any part is covered with fungus, the motor is considered to have failed the test.

4.5.6 Life Test. Using a 360 ohm resistor network, the motor shall be operated 5000 times through a cycle of 40 seconds on and 5 seconds off with an input voltage of 28 $\pm$ 5 volts DC and no load. At the conclusion of each 1000 cycles, the motor shall meet the requirements of 4.5.3.1, 4.5.3.2 and 4.5.3.3, when operated with the negative terminal of the power supply grounded to the motor case.

4.5.7 Vibration Cycling at +160°F. The motor shall be tested per and shall meet the requirements of the vibration tests of this specification except that the unit shall be vibrated while at +160°F.

4.5.8 Vibration Cycling at -65°F. The motor shall be tested per and shall meet the requirements of the vibration tests of this specification except that the unit shall be vibrated while at -65°F.

4.5.9 Shock at -65°F. The motor shall be tested per and shall meet the requirements of the shock tests of this specification except that the unit shall be shocked while at -65°F.

4.5.10 Shock at +160°F. The motor shall be tested per and shall meet the requirements of the shock tests of this specification except that the unit shall be shocked while at +160°F.

4.5.11 Acceleration at +160°F. The motor shall be tested per and shall meet the requirements of the acceleration tests of this specification except that the unit shall be accelerated while at +160°F.

4.5.12 Acceleration at -65°F. The motor shall be tested per and shall meet the requirements of the acceleration tests of this specification except that the unit shall be accelerated while at -65°F.

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4.5.13 High Temperature Storage. The motor shall be subjected to a temperature of +160°F for a period of 100 hours. At the end of 3 hours and the end of 100 hours and while at +160°F, the motor shall meet the requirements of the low voltage, high voltage, coast angle, and the leakage current, Part II tests. The motor shall then be allowed to stabilize at room temperature and then meet the requirements of the low voltage, high voltage, and coast angle tests of this specification.

5. PREPARATION FOR DELIVERY. Preparation of the motor for delivery shall be in accordance with MIL-E-17555.

5.1 Marking. Interior angle exterior containers shall be marked in accordance with MIL-STD-129. The nomenclature shall be:

MOTOR, DIRECT CURRENT FOR CONTROL-MONITORS

6. NOTES

6.1 Intended Use. The motor will be used with the Control-Monitors DCU-116/A, DCU-117/A and DCU-121/A.

6.2 Preproduction Test Samples. It is expected that the contract or purchase order will specify a minimum of three units representative of the production motor will be submitted to the procuring activity for test and evaluation. The invitation for bids, contract or purchase order should specify the point of inspection for these tests.

6.3 Production Delivery. No production motor should be delivered prior to full approval of the preproduction samples.

6.4 Definitions

a. Preproduction Tests. Preproduction tests are those tests performed prior to production on representative samples of the production equipment to demonstrate that the equipment meets the requirements of this specification.

b. Acceptance Tests. Acceptance Tests are those tests accomplished on representative samples of equipment submitted for acceptance under a contract to determine conformance to the requirements of this specification, and to determine that the equipment is equivalent in performance and construction to the approved preproduction samples.

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity *(as indicated on reverse hereof)*.

SPECIFICATION

ORGANIZATION *(Of submitter)*	CITY AND STATE

CONTRACT NO.	QUANTITY OF ITEMS PROCURED	DOLLAR AMOUNT
		$

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES.

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity)*	DATE

DD FORM 1 APR 63 1426 REPLACES NAVSHIPS FORM 4863, WHICH IS OBSOLETE PLATE NO. 15419

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